

PowerNext-AI Screening Submission: Methodology

Problem and data. The workbook contains 1000 labeled training rows and 350 unlabeled test rows. The training targets are Reference_Parameter (regression) and Validity_Label (binary classification). Test_ID is retained only for output and is excluded from modeling. The four sensors and operating variables are used as numeric inputs.

Preprocessing and features. Missing numeric values are median-imputed inside each scikit-learn pipeline, with missingness indicators. Automatically derived features include sensor mean, standard deviation, range, a voltage-current power proxy, and a temperature-duration proxy. Exact duplicates were checked; none were found. IDs are unique. No rows were manually edited or removed.

Reference prediction. An ExtraTreesRegressor (400 trees, minimum leaf size 2, 90% feature subsampling) was selected using five-fold shuffled cross-validation. It captures nonlinear operating-region interactions without requiring an assumed physical equation. Cross-validated MAE was 0.9340, RMSE 2.3522, and R2 0.9516.

Invalid detection. An ExtraTreesClassifier (400 trees, minimum leaf size 3, balanced class weighting) predicts the invalid class from the same observed record-level features. The decision threshold of 0.35 was selected using out-of-fold probabilities; it was not tuned on test data. Five-fold stratified cross-validation produced accuracy 0.9460, invalid precision 0.7985, invalid recall 0.7985, and invalid F1 0.7985. A valid-only regression residual signal was investigated, but it was weaker than the supervised detector and was not used as a standalone rule.

Validation and assumptions. Regression folds were shuffled K-folds; classification folds were stratified. The rows are assumed to be independent because no repeated group key was supplied. The test distribution is assumed to share the workbook's feature semantics. Extreme values are retained and handled by the ensemble rather than deleted. The reported metrics are validation estimates, not hidden-test performance.

Digital-twin automation. In deployment, equipment sensors feed a timestamped acquisition layer. A validation gateway checks schema, units, ranges, missingness, and duplicate events, then applies the locked preprocessing and feature pipeline. The digital twin estimates the current operating state, predicts the reference parameter, and assigns a valid/invalid probability. Confidence, drift, and sensor-health monitors route alerts to operators; confirmed outcomes are stored for periodic recalibration, champion/challenger testing, and controlled retraining.

Input features: Applied_Voltage_kV, Load_Current_A, Ambient_Temperature_C, Test_Duration_min, Sensor_S1, Sensor_S2, Sensor_S3, Sensor_S4